TrES-1b

R-1633

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_210811` · created 2026-08-02T05:59:25+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459437.8373 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.053
Transit depth (Rp/Rs)^2	1.78 +/- 1.42 [%]
Orbital Inclination (inc)	89.9 +/- 1.46
Airmass coefficient 1 (a1)	1.38 +/- 0.16
Airmass coefficient 2 (a2)	-0.23 +/- 0.27
Scatter in the residuals of the lightcurve fit is	12.29 %
Best Comparison Star	#3 - [620, 102]
Optimal Aperture	2.13
Optimal Annulus	4.05
Transit Duration (day)	0.086 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.053 vs 0.1358 (deviation 0.05σ)
Duration fit vs published	0.086 d vs 0.1042 d (deviation 3.54σ)
Mid-time O–C	-5.6 min
Depth SNR	2.4
Residual scatter	12.29 %

Light curve

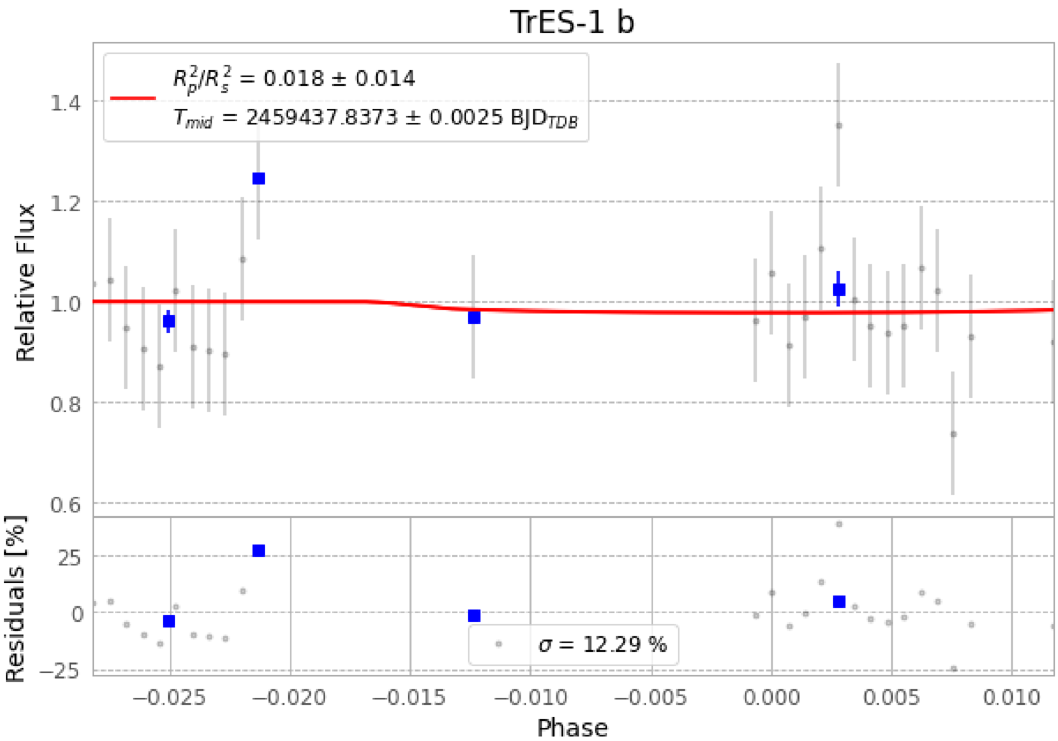

Figure 1 — FinalLightCurve_TrES-1 b_2021-08-10.png (SHA-256 c2a8582a8a4b83f8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 196], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9bbab36644f45e63…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

62 FITS frames (41 MB), `TRES-1210811055712.FITS` → `TRES-1210811090012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-210811.log (6da704f85114…), FinalLightCurve_TrES-1 b_2021-08-10.png (c2a8582a8a4b…), FinalLightCurve_TrES-1 b_2021-08-10.csv (05439b3dfdc0…)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)